Network Science

Lecture 03: Strong and Weak Ties

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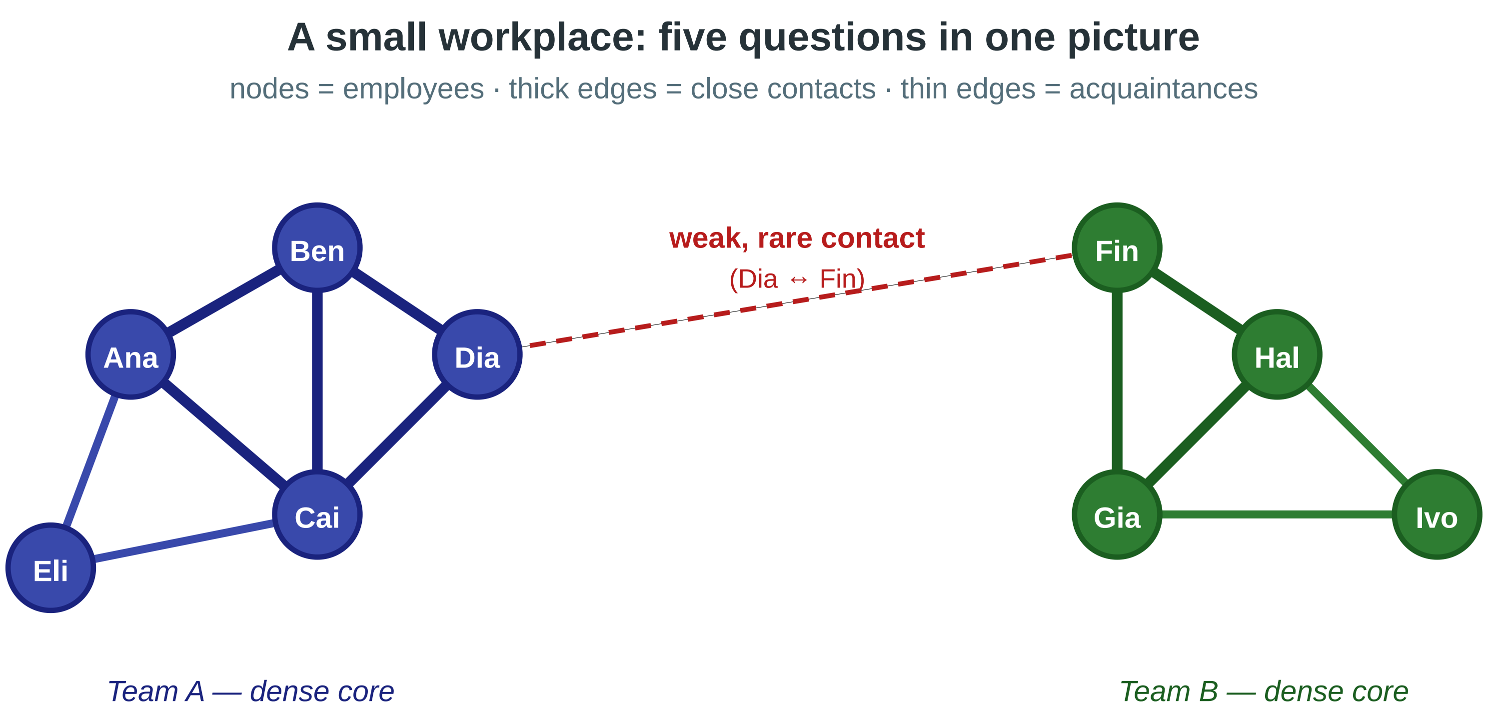
Speaker notes

This lecture is structured around a single tension: the gap between what we would like to **model** in a social network (edge formation, tie strength, information shortcuts) and what we can actually **measure** on real data. Every concept today exists to bridge that gap. Triadic closure and Strong Triadic Closure give us a theory of how friendships form and how ties differ in kind. Exact recovery of that theory from data is computationally intractable, so we introduce structural heuristics — clustering coefficient, neighborhood overlap — that approximate it in polynomial time. The weak-tie theorem then gives a theoretical prediction that those heuristics can test against real-world cell-phone and social-media data.

From Graphs to Social Structure

Last time we built the formal language of graphs. Today we use it to ask a different kind of question: once we have the graph, how do we explain what happens on it?

A Motivating Scenario



Questions we'd like to answer from this picture

① Will Eli and Ben become friends? ② How do we separate close from distant ties?
③ Which single edge controls all information flow between teams? ④ Where would a rumor spread fastest? ⑤ Which tie would we cut to isolate a leak?

Two teams, each tightly connected internally, linked only by a rare contact between Dia and Fin. Thick lines are close daily contacts, the thin dashed line is an occasional one.

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Speaker notes

This figure is the anchor for the entire lecture. Use it to provoke the five questions on the next slide before any definition arrives. Students should be able to point at the picture and name which concept applies where by the end of the session. Each question maps directly to a module: (1) triadic closure, (2) typed edges + STC, (3) bridges and local bridges, (4) weak ties and information flow, (5) robustness under edge removal. Keep this list in mind as the lecture progresses — students should be able to point back to which question is being answered at any moment.

The Catch: Theory vs. Measurement

Each of those five questions has a **theoretical** form and a **measurement** form — and they are not the same.

- The theoretical form asks: *what structure would explain this behavior if we could see everything?*
- The measurement form asks: *what can we actually compute from a real graph where tie strengths, intentions, and labels are hidden?*

Today's lecture lives in the gap between those two.

Today

- triadic closure as a theory of friendship formation
- typed edges and Strong Triadic Closure as an edge-labeling model
- why the exact labeling is NP-hard — and what to do instead
- structural heuristics: clustering coefficient and neighborhood overlap
- the weak-tie theorem and its empirical test at scale

Speaker notes

This is the last slide of the intro. It states the theory-vs-measurement tension explicitly so that students recognize it as the backbone of every subsequent module. The "Today" list is then re-read through that lens: each bullet is either a theoretical construct, a measurable proxy, or a theoretical prediction that links the two.

In a bit more detail

Every question below has a **theoretical** form and a **measurement** form. The lecture deliberately treats them as distinct tasks.

Question	Theoretical construct	Measurable proxy
How do new ties form?	Triadic closure	Triangle density over time
Are ties of different kinds?	Strong / weak edge labeling + STC	Clustering coefficient
Which edges carry novel info?	Local bridge	Neighborhood overlap
Why are bridges weak?	Weak-Tie Theorem	Tie-strength vs. overlap plots
Which ties keep the network together?	Robustness under edge removal	Giant-component size

Speaker notes

This table is the lecture's roadmap in one slide. The left column is what we would *like* to test; the right column is what we can *actually compute* on real data. The lecture's job is to justify each row: why is the proxy a reasonable substitute for the construct, and what does the substitution cost us?

Takeaway

> Network science in this lecture is a loop: we build a theoretical model of edge formation and tie strength, discover that exact inference from data is intractable, replace the model with polynomial-time structural proxies, and test whether the proxies match a theoretical prediction at scale.

Speaker notes

This takeaway is unusual because it describes the *shape* of the lecture, not a single fact. Refer back to it when introducing each new concept — "this is a theoretical construct, not something we can measure directly" or "this is a polynomial-time proxy for the intractable exact problem we just defined".

Triadic Closure

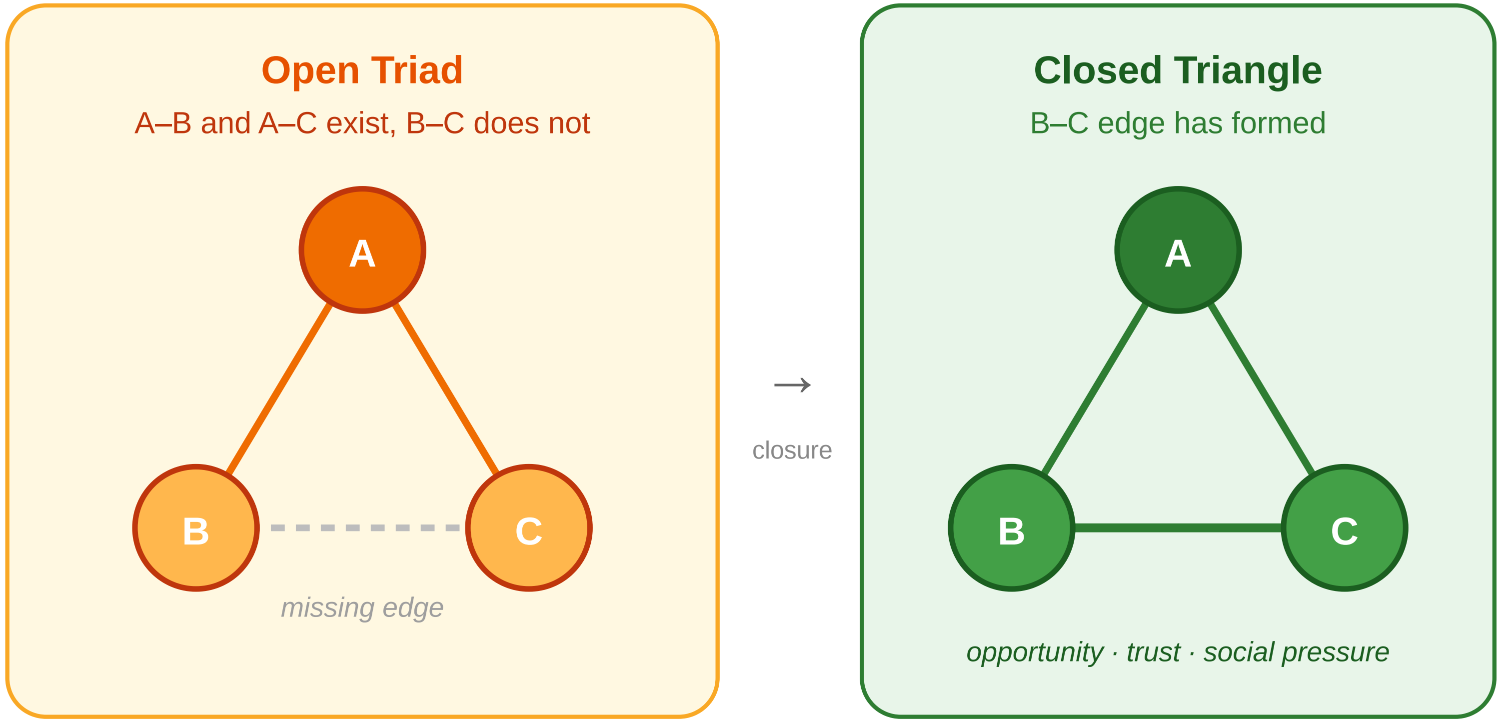
> **Guiding Question:** Why do friends-of-friends so often become linked — and what does that imply for how social graphs grow?

Speaker notes

Triadic closure is the first theoretical construct of the lecture. It is a **generative hypothesis** about how new edges enter the graph. It is not something we measure directly; it is something we *predict* the graph will exhibit if the hypothesis holds.

Open Triads and Closure

Triadic Closure. If a node A has edges to two distinct nodes B and C, the probability that an edge (B, C) forms over time is significantly higher than for two random nodes.



The dashed edge in the open triad is the missing one. Triadic closure predicts it will tend to appear over time, producing the closed triangle on the right.

Speaker notes

Three mechanisms drive closure: **opportunity** (a shared friend creates contexts for B and C to meet), **trust** (a common contact reduces the cost of trusting a stranger), and **incentive** (psychological pressure to resolve cognitive inconsistency in unbalanced triads). Each mechanism is local, but their cumulative effect across a network is the formation of dense clusters separated by relatively few cross-cutting links — exactly the pattern we see in the scenario figure.

Triadic Closure as a Model

Triadic closure is not an observation — it is a claim about how edges appear. As a model, it makes predictions:

- the graph should accumulate triangles over time
- average distances should shrink
- nodes with many neighbors should have increasingly clustered neighborhoods
- long-range ties should become rarer relative to local ones

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Speaker notes

Framing closure as a *model* — something with falsifiable predictions — is the important move. The rest of the lecture gradually tests those predictions on real data. Do neighborhoods really become more clustered? Do long-range ties really end up weak? The answers determine whether the model earns its keep.

Empirical Test: University Email Network

Kossinets & Watts (2006), *Empirical Analysis of an Evolving Social Network*, **Science 311**, 88–90.

They reconstructed the social network of **43 553 students, faculty, and staff** at a large US university from one year of time-stamped email headers, tracking which pairs started exchanging messages over time.

Headline result. Two people who shared **one** mutual email contact were roughly **30 times** more likely to begin emailing each other than two people with no mutual contact. The effect grew monotonically with the number of shared contacts — direct, longitudinal evidence for triadic closure at scale.

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Speaker notes

This study is important for two reasons. First, it is one of the cleanest large-scale confirmations of triadic closure — not inferred from a static snapshot but measured *longitudinally*, by watching new ties appear and asking whether shared contacts predicted them. Second, the effect size is large enough to be unambiguous: 30× is not a statistical nuance, it is a structural regularity. The paper also showed that shared organizational affiliation (same department, same class) independently contributes to closure, so triadic closure is not the whole story — but it is a substantial part of it. See the full paper at https://www.science.org/doi/10.1126/science.1116869 or the author preprint at https://www.albany.edu/~ravi/pdfs/kossinets_watts_2006.pdf.

What the Study Could Not See

Kossinets & Watts observed only one signal per edge: *did an email pass between these two people, yes or no?*

- They could not see whether an edge was a daily collaboration or a one-off administrative ping.
- They could not distinguish a pair who spoke for hours off-platform from a pair who had never met.
- Every edge in the dataset was, by construction, **binary and untyped**.

Speaker notes

This slide is the hinge into the next module. The Kossinets–Watts study demonstrates that closure is real and measurable, but the dataset itself — binary email exchanges — is fundamentally unable to distinguish strong from weak ties. That limitation is not a flaw of the study; it is a limitation of what raw interaction logs can tell us about the social world they come from. The only way forward is to introduce tie strength as an explicit modeling choice rather than hope it will emerge from the data.

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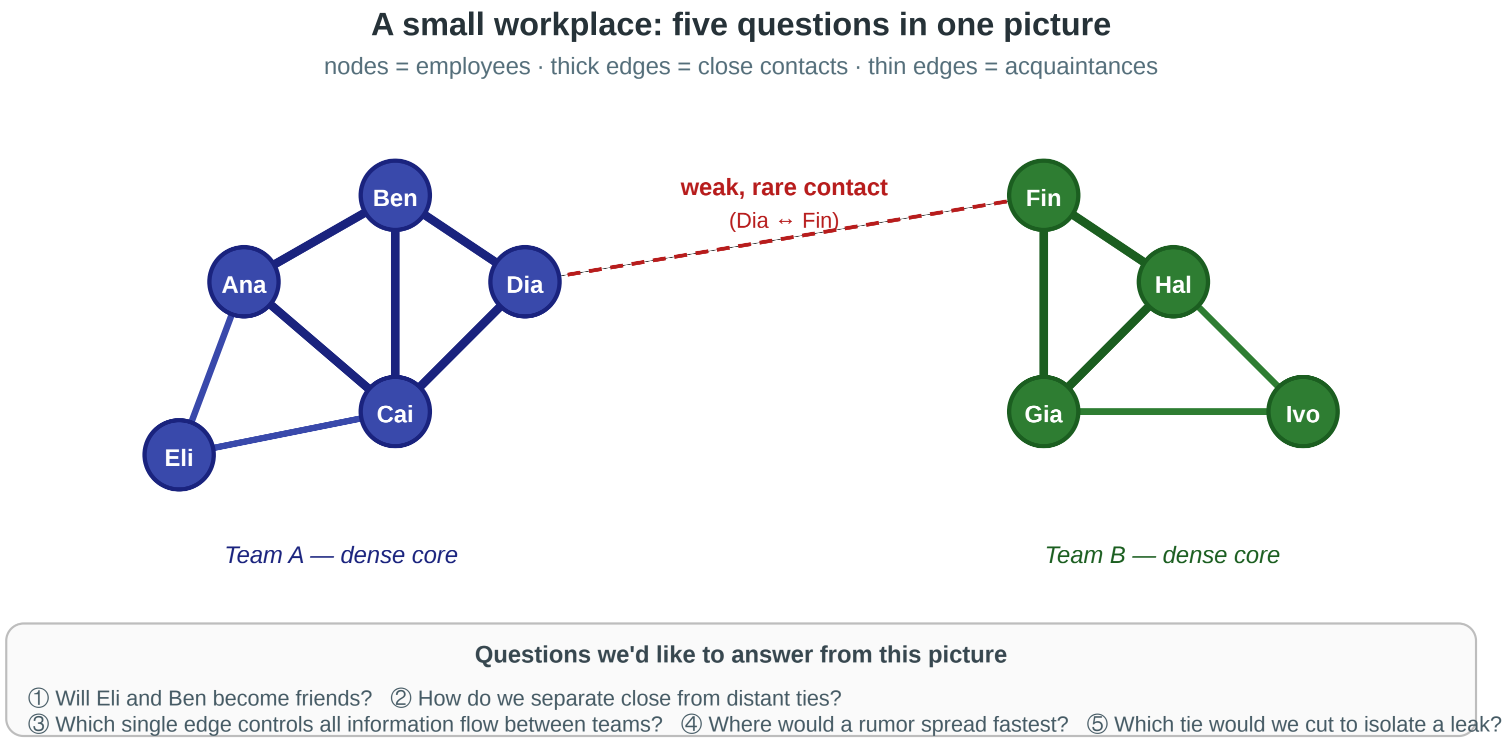
- Every edge in the dataset was, by construction, **binary and untyped**.

Closure is visible in their data. **Tie strength is not.** The next module takes that limitation seriously: if we want to model strong vs. weak ties, we need a formal object — an edge labeling — that the email graph never gave us.

Takeaway

> Triadic closure is a generative theory of edge formation. Kossinets & Watts confirmed it at scale on a university email network — but their binary email data could not distinguish strong from weak ties. To go further, we need to model tie strength explicitly.

Quick Check



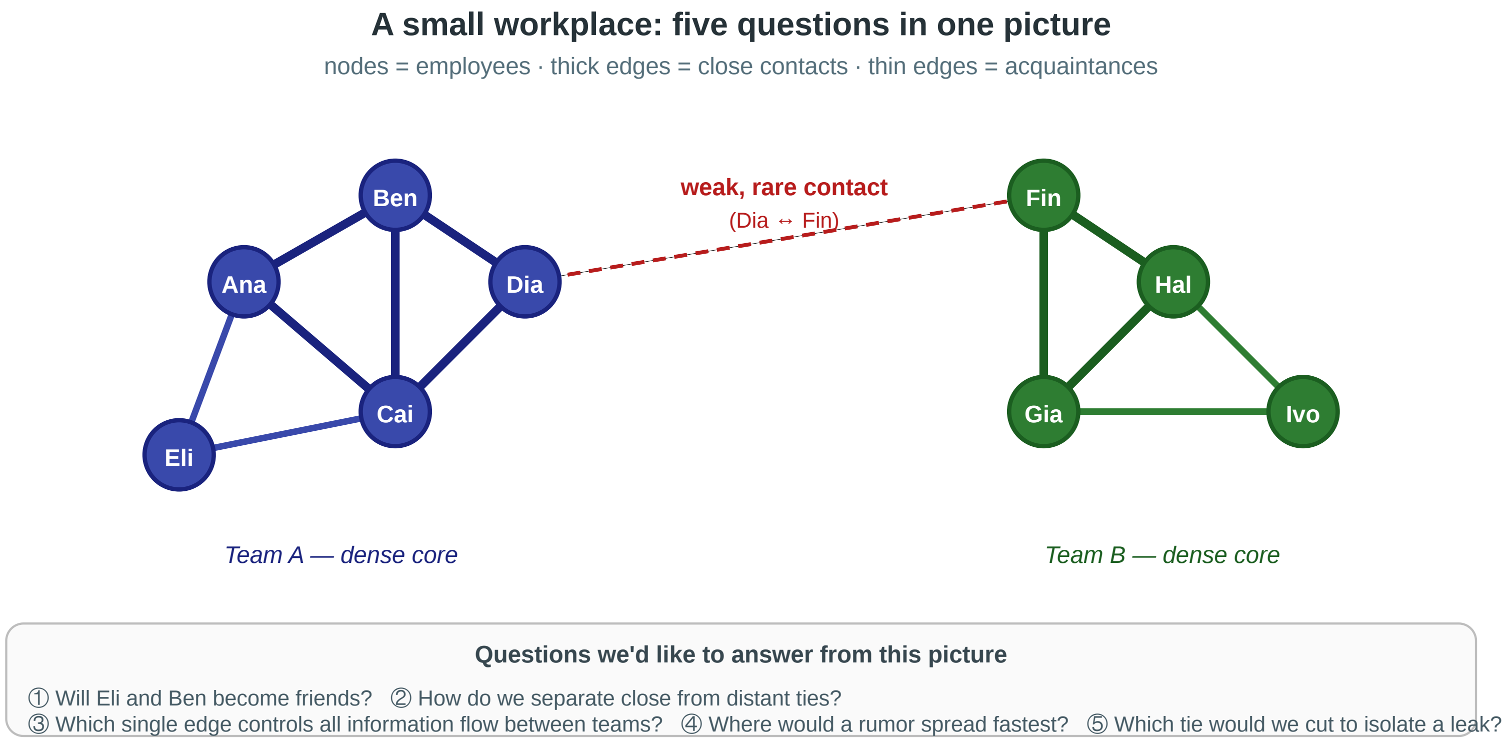
Assume triadic closure is the mechanism shaping this workplace network.

1. Which **missing edge** in the picture would triadic closure most naturally predict to appear next?
2. If that edge does **not** appear, does this already refute the triadic-closure model? Why or why not?

Speaker notes

The point is to make students treat the theory as falsifiable. (1) Closure predicts the missing tie tends to appear. (2) Failure of closure does not refute the model — it predicts a *tendency*, not a certainty — but persistent failure across many such triples would be evidence against the theory or evidence that some other mechanism (enmity, structural barrier) is interfering.

Quick Check



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1. Which **missing edge** in the picture would triadic closure most naturally predict to appear next?
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Solution

1. Triadic closure predicts that the most likely new edge is the one that would **close an open triad**: two people who already share a common contact but are not yet directly connected.

2. A single failed prediction does not refute a probabilistic model. But *persistent* failure across many such triples would indicate either that the closure mechanism is weak in this population or that some competing mechanism — interpersonal conflict, structural

Typed Edges and Strong Triadic Closure

> **Guiding Question:** If ties come in different *kinds*, what constraint does that place on how they coexist around a node?

Speaker notes

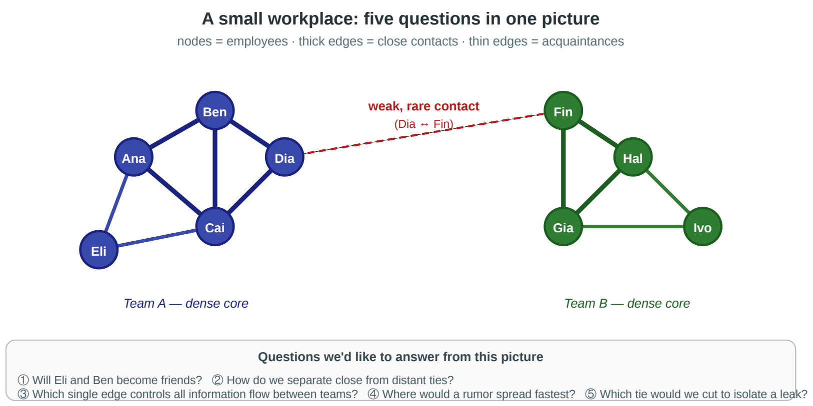
This is the module that earlier versions of the lecture handled implicitly. The key conceptual move is to treat tie strength not as a sociological adjective but as an **edge labeling** — a mathematical object that can be formally defined, constrained, and then *recovered* (or not) from data.

Edges Are Not All the Same

In the scenario, the Ana–Ben edge and the Dia–Fin edge are clearly different:

- Ana–Ben is a daily interaction between people in the same team
- Dia–Fin is a rare, occasional contact between people in different teams

Graph theory's default edge is binary: either present or absent. We now extend it.



Edge labeling. A graph $G = (V, E)$ together with a function

$$\ell : E \to \mathrm{Strong}, \mathrm{Weak}$$

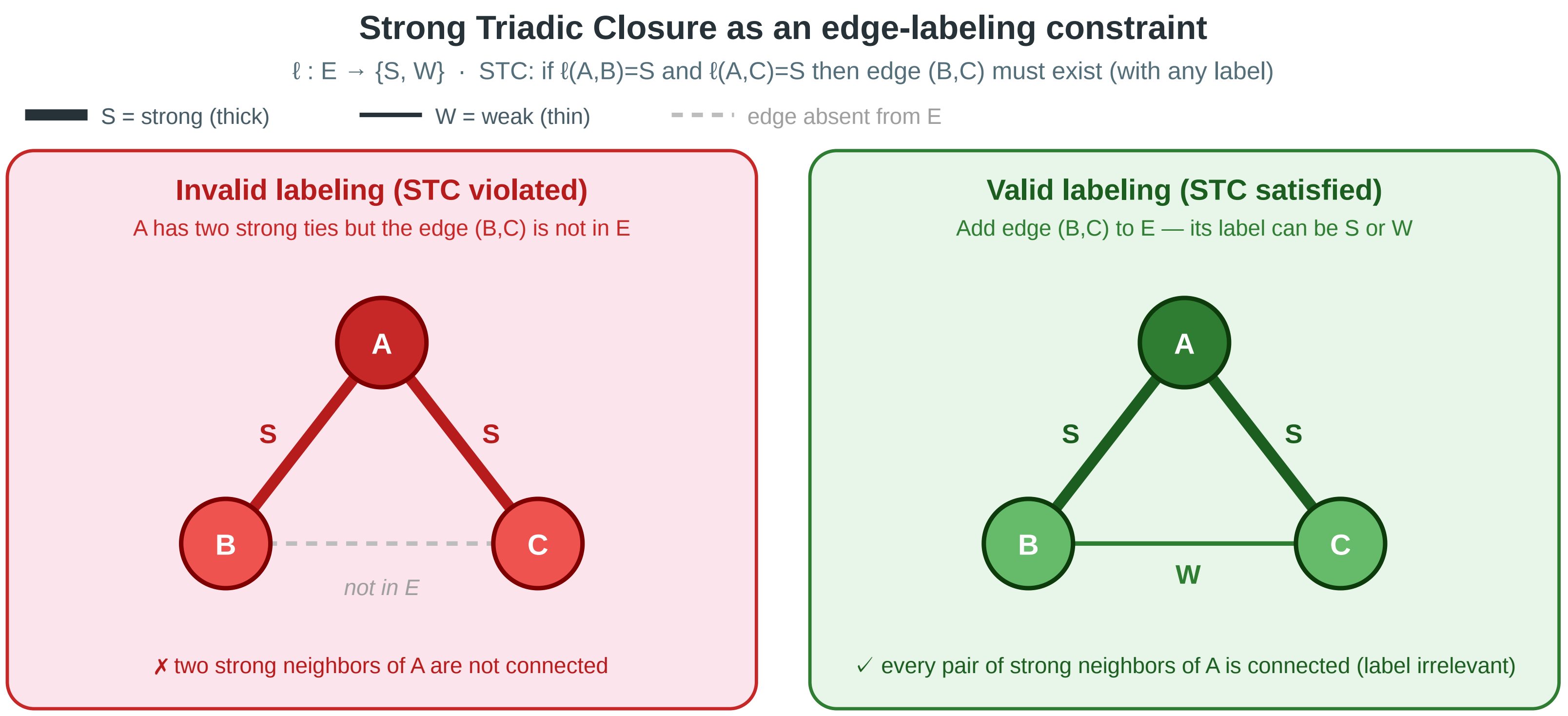
assigning a *type* to every edge. We write $S(v)$ for the set of strong neighbors of v under ℓ.

Speaker notes

This is the crucial definitional step. Instead of saying "some ties are stronger than others" informally, we introduce a formal object: a two-valued labeling function on the edge set. From here on, "strong tie" and "weak tie" mean *an edge with a specific label under a chosen labeling* — not a vague adjective.

Strong Triadic Closure as a Constraint on Labelings

> **Strong Triadic Closure (STC).** A labeling ℓ of G satisfies STC if, for every node v and every pair $u_1, u_2 \in S(v)$ of strong neighbors of v, the edge (u_1, u_2) exists in E — *with any label* (Strong or Weak). STC constrains which pairs must be **connected**, not how the closing edge is labeled.



STC is a *property of a labeling*, not a property of a graph. The same graph can admit valid STC labelings (right) and invalid ones (left). The question "does the graph satisfy STC?" is ill-posed — the right question is "which labelings of this graph satisfy STC?".

Speaker notes

Making STC a property of *labelings* rather than of graphs is the move that clears up the confusion in earlier treatments. A graph by itself can always satisfy STC trivially — label every edge Weak. The interesting question is whether we can label it with *many* strong edges while still satisfying STC. That is the MaxSTC problem.

The Recovery Problem — MaxSTC

In real data we observe the graph G but not the labeling ℓ. Given G, we want to recover the labeling that best explains the observed structure — the one with as many strong edges as possible subject to STC.

> **MaxSTC.** Given $G = (V, E)$, find a labeling ℓ satisfying STC that maximizes $|e \in E : \ell(e) = \text{Strong}|$.

Equivalently, the decision version asks: given G and an integer k, is there an STC-satisfying labeling with at least k strong edges?

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Speaker notes

MaxSTC is the formal bridge from theory to data. It turns the abstract STC constraint into a concrete optimization problem: find the labeling that assigns the most edges to Strong while respecting the constraint. The optimization variant asks for the maximum number of strong edges and a labeling that achieves it. The yes/no decision variant asks: given a threshold k, does there exist an STC-satisfying labeling with at least k strong edges?

Quick complexity vocabulary: **NP** means nondeterministic polynomial time; for our purposes, it means that if someone proposes a labeling, we can check efficiently whether it satisfies STC and has at least k strong edges. Verifying a proposed answer is easy; finding such an answer may still be hard.

NP-hard means at least as hard as the hardest problems in NP: if we had a polynomial-time algorithm for an NP-hard problem like MaxSTC optimization, we could solve a large class of hard problems efficiently. **NP-complete** means both polynomially verifiable (in NP) and NP-hard. For the decision version of MaxSTC, we can verify a proposed labeling efficiently, and Sintos & Tsaparas show the problem is NP-hard; hence the decision version is NP-complete. We do not know whether a polynomial-time algorithm exists for NP-complete problems; finding one would imply $P = NP$.

The NP-hardness result is the hinge of the lecture. Once students accept that the exact labeling cannot be computed on the graphs we care about, the rest of the lecture — clustering coefficient, neighborhood overlap, weak-tie theorem — becomes motivated: these are structural proxies designed to approximate the intractable exact problem in polynomial time. The contrast with restricted tractable classes (cographs, bipartite) is worth mentioning because it tells us the hardness is specific to the unstructured graphs of real social data, not a universal property of the problem.

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Complexity of MaxSTC

> **Result (Sintos & Tsaparas, 2014).** MaxSTC is **NP-hard** as an optimization problem and **NP-complete** as a decision problem on general graphs. Polynomial-time algorithms exist for certain restricted graph classes (e.g., cographs and several bipartite-like structures).

Consequence: on any realistically sized social network, we **cannot** recover the true STC-maximizing labeling exactly.

This is a **modeling problem**, not just an engineering problem. The theoretical construct we care about is computationally out of reach. We need a different strategy.

What We Do Instead

Since we cannot recover the true labeling, we take a different route:

1. Define structural **heuristics** that approximate the "strong"/"weak" character of edges in polynomial time.
2. Derive a theoretical **prediction** from STC (the weak-tie theorem).
3. **Test** the prediction on real data, using the heuristics as proxies for the unobservable labeling.

This is the plan for the next three modules.

Speaker notes

Making the strategy explicit at this point gives students a mental map for everything that follows. They should recognize that the clustering coefficient in the next module exists *specifically* because MaxSTC is intractable. Without the intractability result, the coefficient would look like an arbitrary metric.

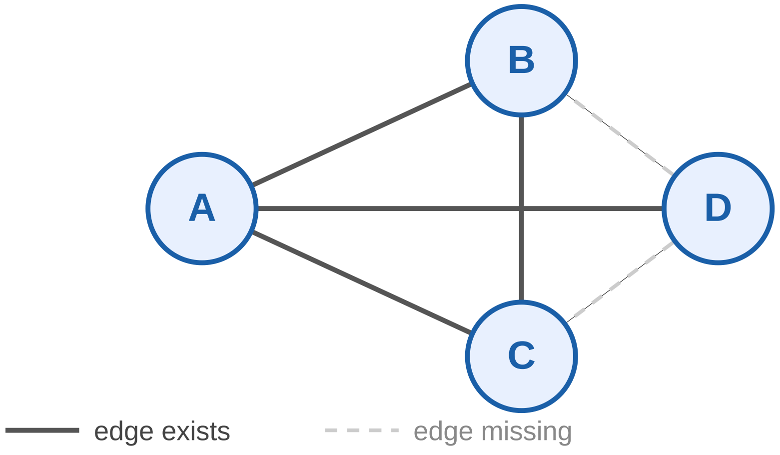
Takeaway

Tie strength is most precisely modeled as an edge labeling, and Strong Triadic Closure is a constraint on such labelings. Recovering the best labeling is NP-hard — so we must approximate the exact theory with polynomial-time structural heuristics.

Quick Check

Consider four nodes A, B, C, D with edges A–B, A–C, A–D, and B–C. No other edges exist (dashed = missing).

1. Find a labeling of this graph that satisfies STC with the maximum number of strong edges.
2. Is the optimal labeling unique?
3. How many possible labellings are there



Speaker notes

(1) The interesting constraint is at node A. If all three of A's ties were Strong, STC would require B–C, B–D, and C–D — but B–D and C–D are missing. So at most two of A's ties can be Strong, and they must be the pair whose endpoints are connected: A–B and A–C (since B–C exists). The edge B–C itself can be strong (no constraint violated). A–D must be weak. Maximum is 3 strong edges. (2) Not unique: A–D could be strong if *all* of A's other ties were weak, giving only 1 strong edge — but that's not the maximum. The maximum labeling (3 strong edges: A–B, A–C, B–C; A–D weak) is unique in this small graph.

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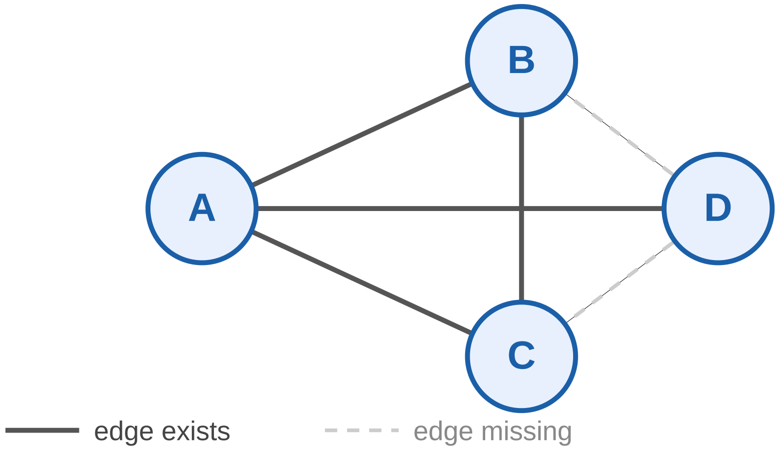


Solution

1 +3. Optimal labeling. We have $2^4 = 16$ possible labellings. At node A, making all three ties strong would force B–D, C–D, and B–C to exist. Only B–C exists, so at most *two* of A's ties can be strong — and they must be the two whose endpoints are actually connected: A–B and A–C. The edge B–C can be strong without further constraint. So:

$$\ell(A{-}B) = \ell(A{-}C) = \ell(B{-}C) = S, \quad \ell(A{-}D) = W$$

gives **three** strong edges — the maximum.

2. Uniqueness. In this small graph, yes. On larger graphs, MaxSTC can have exponentially many optimal labelings — another reason the problem is hard.

Structural Heuristics for Tie Strength

> **Guiding Question:** Without access to the true edge labels, how can we estimate which parts of a network behave like "strong, embedded" regions and which behave like "weak, bridging" ones?

Speaker notes

Now the clustering coefficient finally has a reason to exist. It is not an arbitrary metric — it is a polynomial-time **structural heuristic** designed to capture the feature of the network that STC would reveal if we could solve MaxSTC exactly.

Clustering Coefficient

Local clustering coefficient. For a node i with degree k_i:

$$C_i = \frac{\text{edges among } i\text{'s neighbors}}{\binom{k_i}{2}}$$

$C_i \in [0, 1]$. $C_i = 1$ means i's neighborhood is a complete clique; $C_i = 0$ means none of i's neighbors are linked.

Why this works as a heuristic. Under STC, if v has many strong ties, those ties' endpoints must all be mutually connected — i.e. v's neighborhood is dense. Conversely, a high C_v indicates a neighborhood that *could* be labeled mostly Strong without violating STC. C_v is thus a lower-bound signal for "how strong-embedded is v".

Speaker notes

The clustering coefficient is computable in $O(k^2)$ per node — polynomial and cheap — while MaxSTC is NP-hard. The price of the heuristic is that it confounds two things: it is high both when the node is genuinely strong-embedded and when it happens to sit in a dense region for unrelated reasons. This is the typical tradeoff of intractability: we lose precision to gain computability.

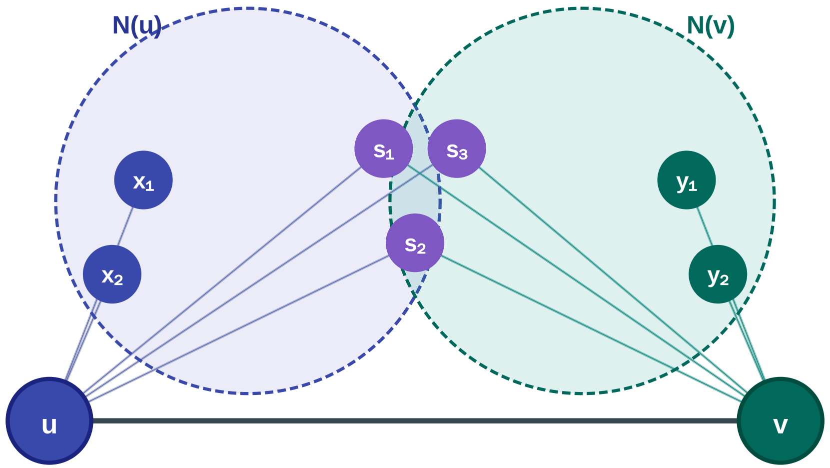
Neighborhood Overlap

Neighborhood overlap. For an edge (u, v):

$$O(u,v) = \frac{|N(u) \cap N(v)|}{|N(u) \cup N(v)|}$$

where $N(u)$ and $N(v)$ exclude each other.

- $O = 0$ — pure local bridge: no shared neighbors
- $O = 1$ — edge inside a complete clique
- Jaccard similarity of the two neighbor sets



Speaker notes

Neighborhood overlap is the edge-level analogue of the clustering coefficient: it measures how "inside" the edge lies. An edge with many shared neighbors sits deep in a dense region and is a reasonable candidate for a strong label; an edge with no shared neighbors is a local bridge and a reasonable candidate for weak. Like the clustering coefficient, it is computable in polynomial time, and like the clustering coefficient, it confounds structural and sociological strength. Both heuristics are approximations to the intractable exact labeling — but they are approximations we can actually compute at the scale of millions of nodes.

Why These Heuristics Instead of Exact Labels

	Exact MaxSTC	Clustering coeff.	Overlap
Scope	whole graph	per node	per edge
Complexity	NP-hard	$O(k^2)$ per node	$O(k)$ per edge
Captures	true tie type	neighborhood density	edge embeddedness
Feasible on real graphs?	No	Yes	Yes

Here k is the **degree** of the node (or of an edge's endpoint) — i.e. its number of neighbors. Clustering coefficient checks all $\binom{k}{2}$ pairs of a node's neighbors; overlap intersects the two endpoints' neighbor sets.

Speaker notes

This table makes the modeling/measurement split concrete. The exact problem is what we want; the heuristics are what we can have. The rest of the lecture (bridges, weak-tie theorem, empirical tests) all operate on the heuristics, not the exact labels, because the exact labels are unavailable.

Takeaway

Clustering coefficient and neighborhood overlap exist because MaxSTC is intractable. They are polynomial-time structural proxies for a theoretical object we cannot compute — trading precision for feasibility on real-world networks.

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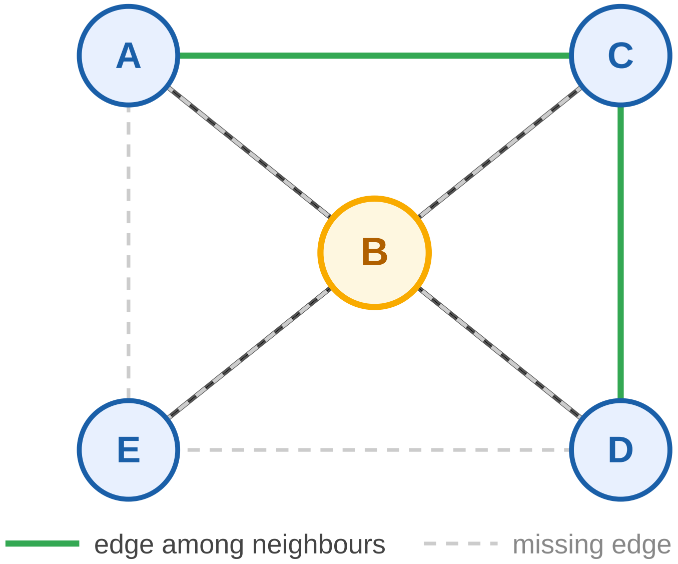
Quick Check

Node B has 4 neighbors: A, C, D, E. Among B's neighbors, exactly 2 edges exist (green): (A, C) and (C, D).

1. Compute C_B.
2. Under STC, what is the maximum number of strong ties B can have?
3. What does C_B tell you about B's structural role that STC alone does not?



Speaker notes

(1) $C_B = 2/6 = 1/3$. (2) For a pair of B's neighbors to both receive strong labels from B, they must be connected. The pairs (A, C) and (C, D) are connected. So B can make strong ties to A, C or to C, D — three strong ties would require all pairs among the selected triple to exist, which they don't. Max strong ties from B = 2. (3) The clustering coefficient gives a continuous ranking — B sits in a moderately sparse neighborhood — while STC only tells us a hard upper bound on the number of strong ties. The heuristic gives more nuance at the cost of exactness.

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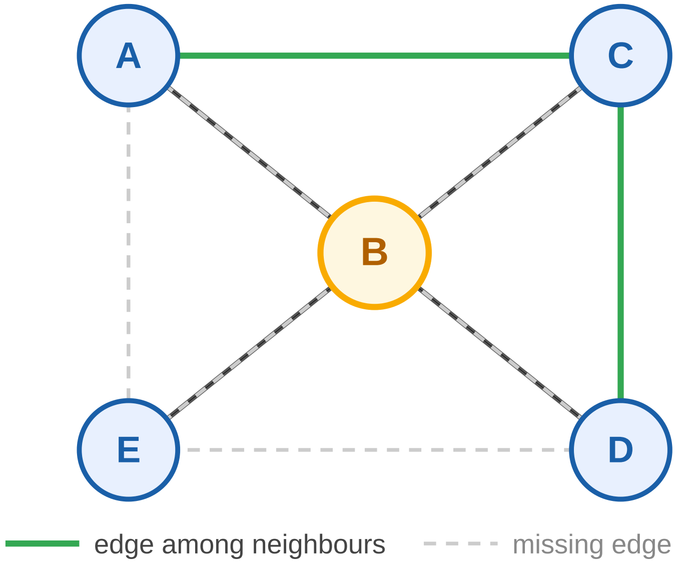


Quick Check — Solution

1. $\binom{4}{2} = 6$ possible edges, 2 actual $\Rightarrow C_B = 1/3$.

2. Maximum strong ties from B is **2**. The only pairs of B's neighbors connected by an edge are (A, C) and (C, D); a third strong tie would force a currently missing edge (e.g. A–D), violating STC.

3. The clustering coefficient is a *continuous* structural measurement — $C_B = 1/3$ tells us B sits in a moderately sparse neighborhood, suggesting a potential brokerage position between the sub-clusters A, C and D. STC by itself gives only a hard upper bound on the

Bridges, Local Bridges, and the Weak-Tie Theorem

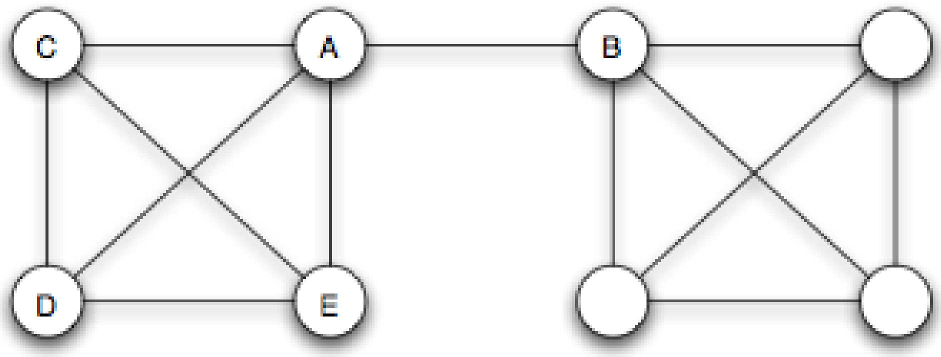
> **Guiding Question:** Which edges are structurally most important for keeping information flowing across a network — and can we predict from theory which ones those will be?

Speaker notes

This module derives the theoretical prediction that lets us test the model against real data. The weak-tie theorem says: if a node satisfies STC and has at least two strong ties, then any local bridge incident to that node cannot be one of those strong ties. That prediction is testable using the heuristics from the previous module.

Bridge

Bridge. An edge $e \in E$ is a *bridge* if removing it increases the number of connected components. Equivalently: e lies on no cycle.



Source: Easley & Kleinberg 2010

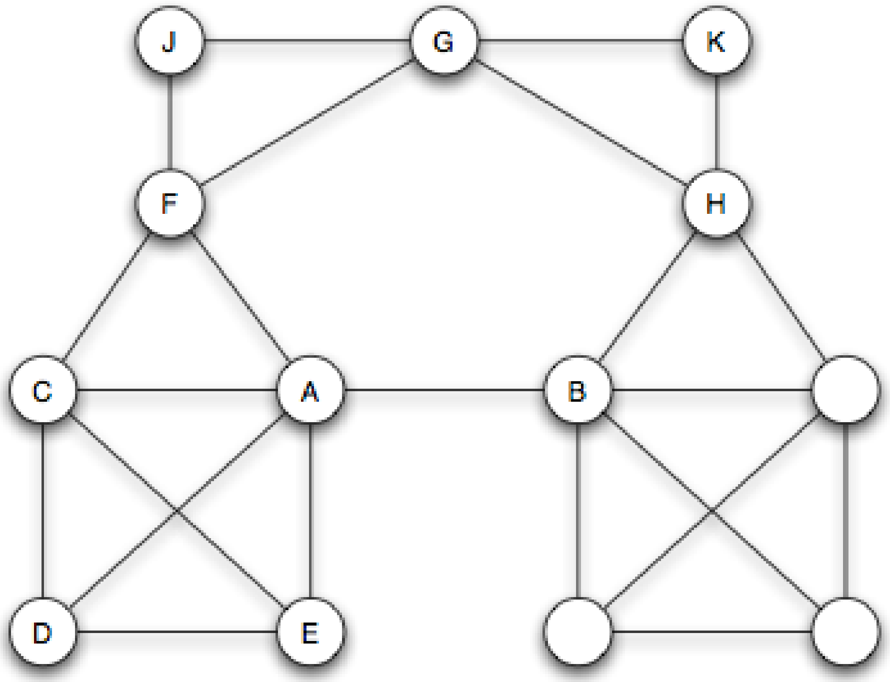
Removing the highlighted edge splits the graph in two. The endpoints' distance becomes infinite — they can no longer reach each other through any other route.

Speaker notes

A true bridge is the strongest possible structural claim about an edge. In real networks, true bridges are rare because alternative routes almost always exist. This motivates the weaker but more useful concept of a *local bridge*.

Local Bridge

Local bridge. An edge (u, v) is a *local bridge* if u and v have no neighbors in common: $N(u) \cap N(v) = \emptyset$. Equivalently: $O(u, v) = 0$.



Source: Easley & Kleinberg 2010

A local bridge does not need to disconnect the graph — its endpoints may still reach each other via a long detour. Note that *local bridge* is already expressible in terms of the neighborhood-overlap heuristic: an edge with $O = 0$ is exactly a local bridge.

Speaker notes

Pointing out that local bridges are exactly the $O = 0$ edges is the link between the theoretical and measurement sides of the lecture. The construct ("local bridge") is theoretical; the quantity (O) is measurable. The equivalence at $O = 0$ extends smoothly into the continuous case: low-O edges are "almost-local-bridges", and they behave like bridges for information flow.

The Weak-Tie Theorem

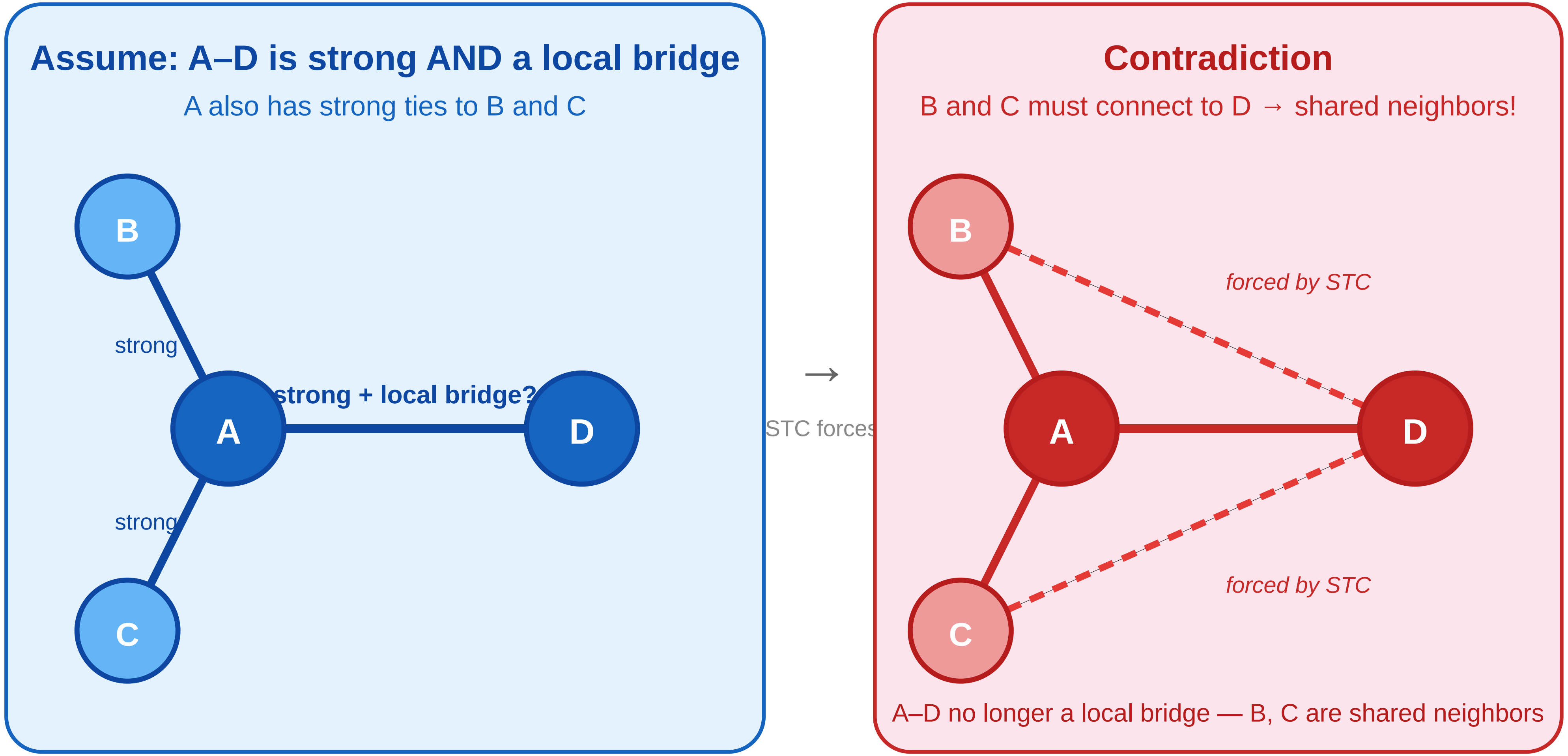
> **Theorem (Granovetter).** If a node a satisfies Strong Triadic Closure and is incident to at least two strong ties, then any local bridge incident to a must be labeled Weak.

This is the theoretical prediction we will test. It links the labeling model (STC, tie types) to a measurable structural property (local bridges) via a local if–then statement.

Speaker notes

State the conditional form explicitly: the theorem is local to a node. It does not say that every local bridge in every graph is weak. It says that once node a satisfies STC and has at least two strong ties, a local bridge adjacent to a cannot be strong; otherwise STC would force a shared neighbor, contradicting the local-bridge condition. Real networks rarely satisfy STC exactly — they violate it locally. The theorem still tells us what we should see *on average*: edges with low neighborhood overlap should tend to be weak. That averaged prediction is what we will test on cell-phone data.

Proof Sketch by Contradiction



Assume A–D is both **strong** and a **local bridge**. Since A has another strong tie, say A–B, STC forces B–D to exist. But then B is a shared neighbor of A and D — destroying the local-bridge condition. Contradiction.

Speaker notes

The proof is one of the cleanest in network science. A single application of STC is enough to derive a contradiction. The structural lesson is that strong ties drag their endpoints into shared social environments, and shared environments destroy the bridge property by definition.

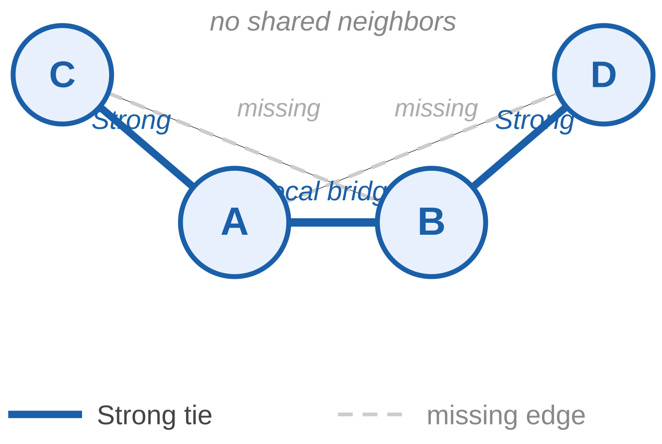
Takeaway

> For nodes satisfying STC with at least two strong ties, incident local bridges must be weak ties. This is the theoretical prediction that connects our unobservable labeling model to the measurable neighborhood-overlap heuristic — and it is the prediction we now test against real data.

Quick Check

A–C and B–D are labeled **Strong**. The edge A–B is the only link between the two sides; dashed edges are missing.

1. Is A–B a local bridge?
2. Could A–B also be labeled **Strong** under STC?



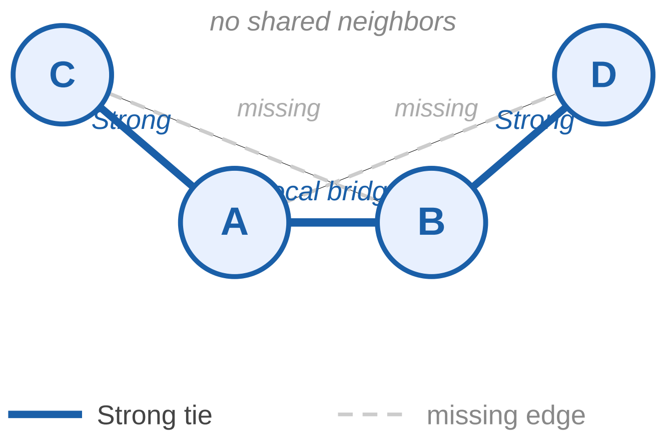
Speaker notes

(1) Yes — A and B have no shared neighbor. C is adjacent to A but not B; D is adjacent to B but not A. (2) If A–B were Strong, then the pair of strong ties A–B and A–C would force B–C; likewise A–B and B–D would force A–D. Since both cross edges are missing, the bridge cannot be Strong under STC. This quick check is the theorem in miniature.

Quick Check

A–C and B–D are labeled **Strong**. The edge A–B is the only link between the two sides; dashed edges are missing.

1. Is A–B a local bridge?
2. Could A–B also be labeled **Strong** under STC?



Quick Check — Solution

1. Yes. $N(A) = B, C$ and $N(B) = A, D$, so A and B have no common neighbor.

2. No. If A–B were Strong, then STC at A would require B–C, and STC at B would require A–D. Both are missing. Therefore, any STC-satisfying labeling that keeps A–C and B–D strong must label the local bridge A–B as **Weak**.

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Empirical Validation at Scale

Guiding Question: Does the theoretical prediction — low overlap correlates with low tie strength — survive contact with real-world data at scale?

Speaker notes

Everything in the previous modules was setup for this one. The weak-tie theorem predicts that, under STC, low-overlap edges should be weak. We now test whether that prediction holds on massive real datasets where neither STC nor the true labeling is directly observable — only structural proxies.

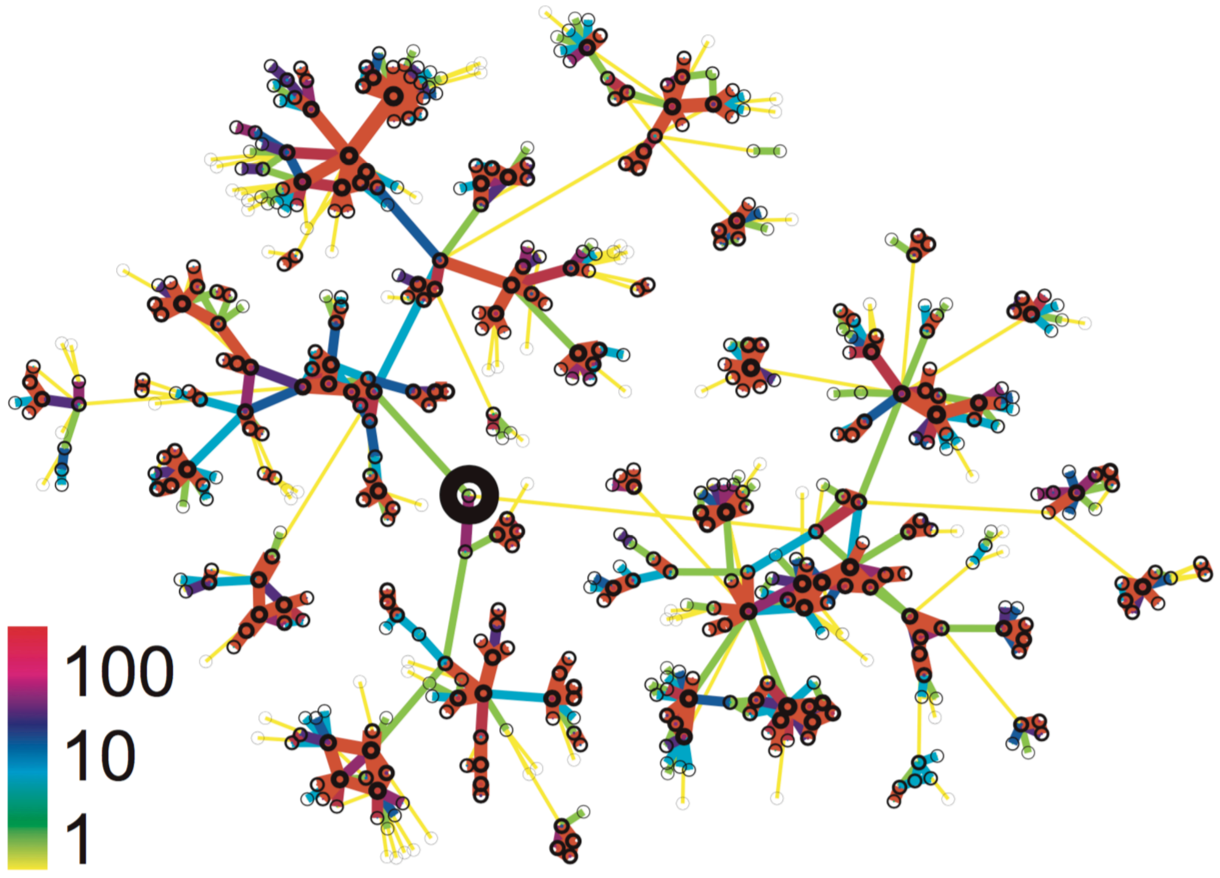
Study 1: Cell-Phone Network (Onnela et al. 2007)

Source: Onnela et al. 2007

Question	Does tie strength correlate with structural embeddedness in a real large-scale network?
Data	Nationwide mobile call logs — 18 months, ~7 M users, one operator
Graph	Undirected weighted: nodes = phone numbers, edge weight = cumulative call duration
Measures	Tie-strength proxy: call duration (percentile); structural proxy: neighborhood overlap $O(u, v)$; global: giant-component size
Prediction	Weak-tie theorem: low-overlap edges are weak, high-overlap edges are strong — monotone relationship

> **Limitations:** Call duration misses topic, reciprocity, and face-to-face contact. Single country and operator limits generalizability. No ground-truth labels to validate the proxy.

Local ego-network of one subscriber. Dense cluster of high-duration contacts; peripheral nodes are reached only by low-duration, low-overlap ties.



Source: Onnela et al. 2007

Speaker notes

The operationalization matters. Call duration is not tie strength in the sociological sense — it captures one dimension (sustained communication) while ignoring others (topic, reciprocity, face-to-face history). But this is exactly the modeling/measurement gap the lecture is about: the true strength is inaccessible, the proxy is computable, and the question is whether the proxy behaves as the theory predicts. The figure shows that even at the single-node level, the pattern is visible: dense cluster of long-duration contacts, sparse periphery of short-duration contacts.

Study 1 — Result 1: Overlap vs. Tie Strength

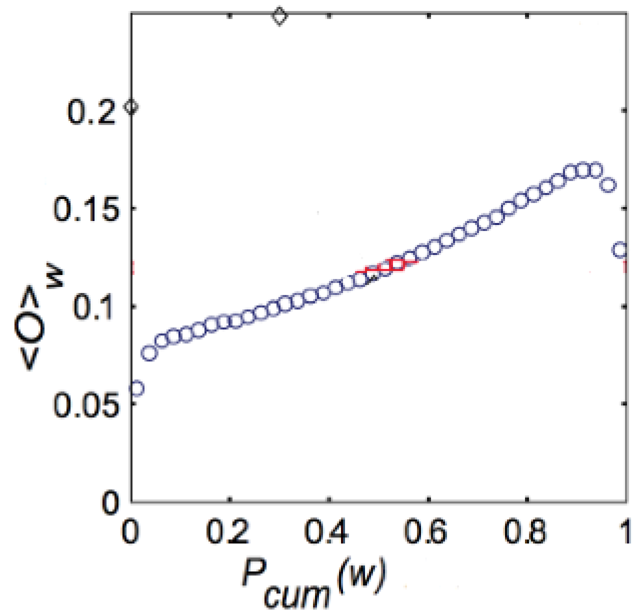
Source: Onnela et al. 2007

Question	Do low-overlap edges tend to be weak and high-overlap edges strong?
Procedure	Bin all edges by call-duration percentile; compute mean $O(u, v)$ per bin
X-axis	Tie-strength percentile (weakest left → strongest right)
Y-axis	Mean neighborhood overlap per bin

Outcome: Clear, smooth monotone relationship — weakest ties cluster near $O \approx 0$; strongest ties accumulate at high O.

Interpretation: Two independently computable proxies align exactly as the weak-tie theorem predicts — not a threshold but a continuous gradient.

$\langle O \rangle$ is the **average neighborhood overlap** of edges in a tie-strength bin. $P_{\text{cum}}(w)$ is the **cumulative tie-strength percentile:** values near 0 are the weakest call-duration ties, values near 1 are the strongest.



Source: Onnela et al. 2007

Limitations: Both axes are proxies. Geographic proximity could independently inflate both call duration and overlap, confounding the tie-type interpretation.

Speaker notes

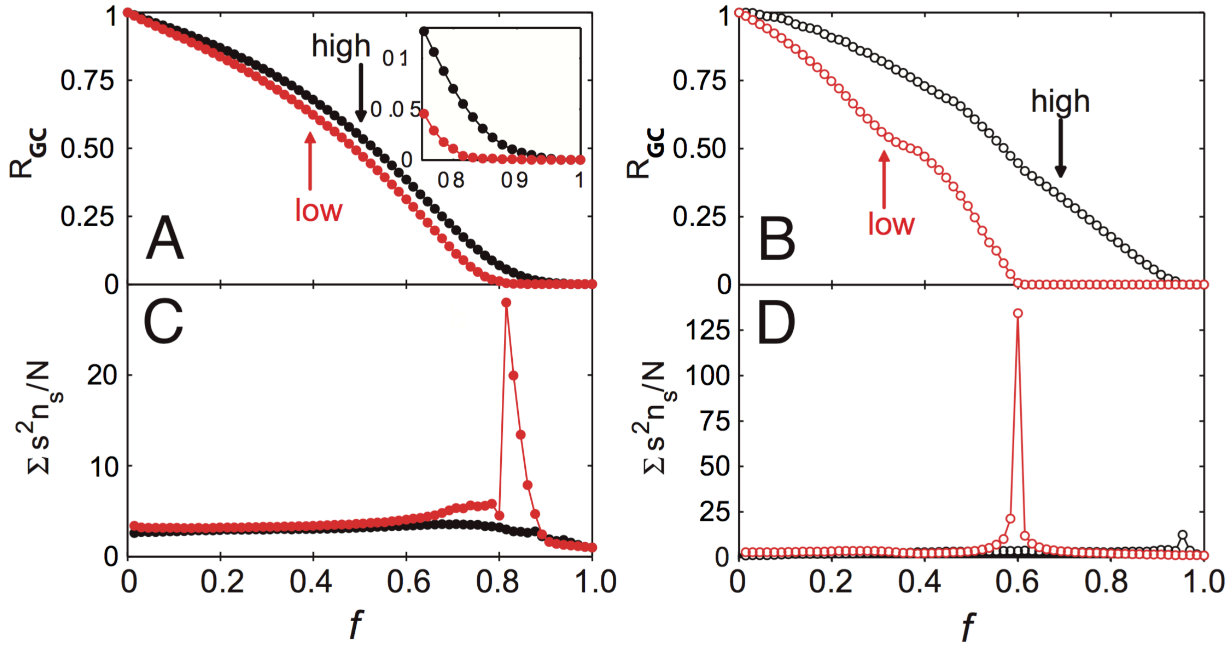
This plot is the empirical heart of the lecture. The fact that the relationship is smooth and monotonic, not just a threshold effect, is evidence that the underlying mechanism scales gracefully from idealized triads to realistic graphs. Crucially, neither axis directly measures what the theory is about — call duration is not "tie strength" and overlap is not "local-bridgeness" — yet the proxies co-vary exactly as predicted. That robustness under proxy substitution is evidence for the generality of the model.

Study 1 — Result 2: Knockout Experiment

Source: Onnela et al. 2007

Question	Are weak ties structurally load-bearing or peripheral?
Procedure	Remove edges in two orders: weakest-first (low call duration) vs. strongest-first
Measure	Giant connected component size after each removal step
Condition A	Weakest ties removed first
Condition B	Strongest ties removed first (control)

Axis notation: f is the fraction of removed edges. R_{GC} is the fraction of all nodes still in the giant connected component. $\sum_s s^2 n_s / N$ summarizes fragmentation outside the giant component: n_s is the number of components of size s, and peaks mark the point where the network breaks apart.



Source: Onnela et al. 2007

Outcome: Weakest-first removal fragments the giant component far faster than strongest-first — weak ties are the network's connective tissue.

Interpretation: Confirms the structural role the theorem assigns: local bridges (weak ties) hold the global network together.

Limitations: Removal is a static simulation, not a real intervention. Giant-component size is coarse — betweenness or average path length might show finer differences. Removal order uses call duration as proxy, not true tie labels.

Speaker notes

This is the knockout test. If weak ties were sociologically marginal *and* structurally redundant, removing them would have little effect. The opposite happens. The weak ties are the connective tissue — exactly the role the theorem assigns them. Together with the overlap plot, this gives two independent empirical confirmations of the theoretical prediction: one correlational (overlap vs. strength), one causal-by-removal (structural impact of deleting weak ties).

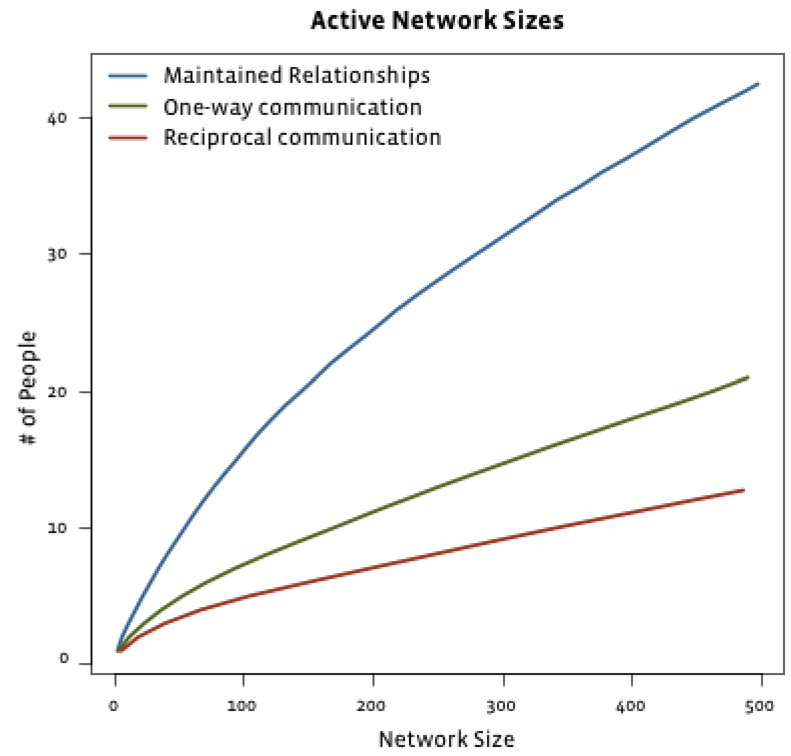
The lower panels use $\sum_s s^2 n_s / N$ as a fragmentation signal, where n_s is the number of connected components of size s outside the giant component and N is the total number of nodes. Intuitively, this is the expected finite-component size reached by following a random node, up to the exact convention used for excluding the giant component. The s^2 appears because components are weighted by both how many such components exist and how many nodes they contain: a component of size s contributes s possible starting nodes, and each such start lands in a component of size s, giving $s \cdot s = s^2$. This makes a few large fragments count much more than many isolated singletons. The value is nonnegative; with this normalization it can be larger than 1 and, in the extreme case where all nodes outside the giant component form one large finite component, it is on the order of N. In practice we read it comparatively: it peaks near the percolation point, when the giant component has just broken and large finite fragments appear.

Study 2: Facebook — Maintained vs. Total Friendships

Source: Gon\ccalves et al. 2011

Question	Do total online friends and actively maintained ties scale the same way?
Data	Facebook friendship and interaction logs across users with varying friend counts
Measure A	Total declared friends (any structural link)
Measure B	Maintained friends (reciprocal messages / comments / wall posts)

Outcome: Total friends grow without bound. Maintained ties plateau around ~150 — consistent with Dunbar's cognitive limit on strong-tie capacity.
Interpretation: Platforms remove the cost of weak ties but leave the cognitive budget for strong ties unchanged. The weak/strong distinction is amplified, not collapsed.



Source: Easley & Kleinberg 2010

> **Limitations:** "Maintained" captures only on-platform interactions; offline contact is invisible. The plateau may partly reflect News Feed ranking rather than purely cognitive limits. Dunbar's ~150 number is itself debated.

Speaker notes

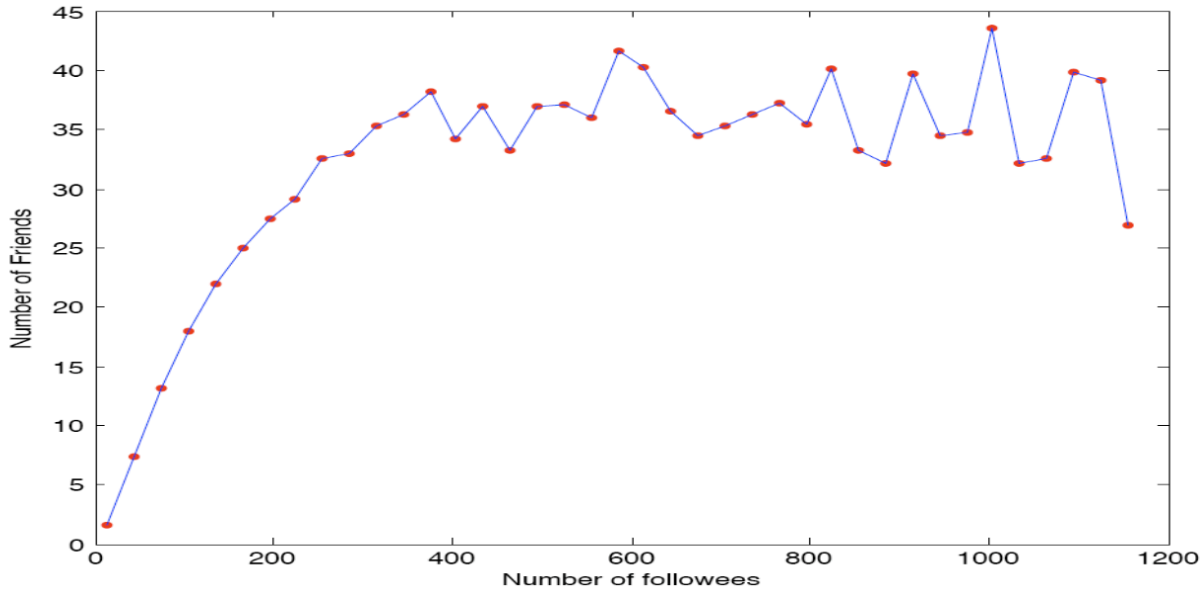
The Facebook result is sometimes cited as evidence that social media "flattens" social structure — everyone is equally reachable. The maintained-vs-total distinction refutes that reading: the flat part of the curve is weak-tie accumulation, not strong-tie expansion. The plateau around 150 replicates a finding from offline social networks (Dunbar 1992) in a completely different medium and scale, suggesting the cognitive bound is robust across communication technology.

Study 3: Twitter — Followers vs. Interacting Friends

Source: Huberman et al. 2008

Question	Does follower count predict active interaction count on an asymmetric platform?
Study	Huberman, Romero & Wu, arXiv:0812.1045, 2008: "Social networks that matter: Twitter under the microscope"
Data	Twitter activity logs
Measure A	Followers: any account that subscribed (passive, asymmetric)
Measure B	Active friends: accounts the user sent ≥ 2 directed messages to

Outcome: Follower count scales freely (power-law). Active interacting friends plateau at a small number, independent of follower count.
Interpretation: Same pattern as Study 2, different platform. Following is cheap; active communication is cognitively bounded. Weak ties scale with affordances; strong ties do not.



Source: Huberman et al. 2008

Limitations: Twitter following is often parasocial — it does not map cleanly onto "weak tie." The ≥ 2-message threshold is arbitrary. Algorithmic timelines shape interaction independently of genuine tie strength.

Speaker notes

Social media is sometimes claimed to "collapse" the distinction between weak and strong ties by making everyone reachable. The empirical evidence says the opposite: platforms remove the cost of maintaining weak ties, so the weak-tie periphery expands dramatically, while the strong-tie core remains bounded by the same cognitive limits as before. The structure is stretched, not flattened. Together with the Facebook result, this gives convergent evidence from two independent platforms with different interaction models (undirected friendship vs. directed following).

Study 4: Facebook Information Diffusion

Source: Bakshy et al. 2012

Question	Who spreads web links on Facebook: strong ties or weak ties?
Study	Bakshy, Rosenn, Marlow & Adamic, WWW 2012
Design	Field experiment randomizing exposure to friends' URL-sharing signals in News Feed
Scale	253 million subjects
Tie strength	Four prior-interaction counts: messages, comments, photo co-tags, shared comment threads
Weak tie cutoff	Aggregate comparison: $k = 0$ prior interactions = weak; $k \geq 1$ = strong

This is a direct online test of Granovetter's idea: tie strength is measured, exposure is experimentally varied, and the outcome is actual sharing behavior.

Finding: Strong ties are more influential *per exposure*, but weak ties generate most information diffusion overall because they are far more numerous.

Novelty: Weak-tie exposure has the largest relative effect because it surfaces links users were less likely to discover independently.

Interpretation: Strong ties transmit well locally; weak ties carry diverse information across the broader network.

Limitations: Tie strength is inferred from Facebook interaction logs, not self-reported relationship quality. The platform's News Feed ranking is part of the treatment environment.

Speaker notes

Bakshy, Rosenn, Marlow, and Adamic measured tie strength from prior Facebook interaction during the three months before the experiment: private messages received, comments received, co-appearances in the same photo, and both users commenting on the same post thread. They then used randomized variation in News Feed exposure to estimate whether seeing a friend share a URL causally increased the probability that a user later shared the same URL. This matters because simple correlation cannot distinguish influence from homophily: friends may share the same link because they independently like the same sources.

The important distinction is individual effect versus collective contribution. A strong tie is more influential conditional on exposure: if a close friend shares a link and it appears in your feed, you are more likely to reshare it than if a distant contact shares it. But most users have far more weak ties than strong ties. For the aggregate comparison, the paper uses a deliberately generous cutoff for strong ties: if there was at least one prior interaction on the chosen metric ($k \geq 1$), the tie is counted as strong; if there were zero prior interactions ($k = 0$), it is counted as weak. Even with that generous definition, weak ties account for the majority of information diffusion because they are so abundant.

They also found that weak-tie exposure has high novelty: without the weak-tie exposure, the user was less likely to encounter or share that URL independently. This maps cleanly onto Granovetter. Strong ties are high-bandwidth local channels; they transmit well inside dense clusters. Weak ties are individually weaker but collectively important because they are numerous and connect users to less redundant information sources.

Closing the Loop

We set out to test a theoretical model (typed edges + STC) on data where the true labels are unobservable and the exact recovery problem is NP-hard. The strategy was:

1. Define polynomial-time structural heuristics (clustering coefficient, overlap)
2. Derive a prediction from the theory (weak-tie theorem)
3. Measure the heuristics on real data and compare

The Onnela and social-media results close the loop: the heuristics behave exactly as the theorem predicts, and the same structural logic explains why weak ties carry novel information online. The model earns its keep even though we could never compute its exact solution.

Speaker notes

This is the moment to refer back to the table from Module 1. Every row has now been addressed — the theoretical constructs on the left are connected to the measurable proxies on the right, and the empirical results validate the connection. This is what a successful model-to-data loop looks like in network science.

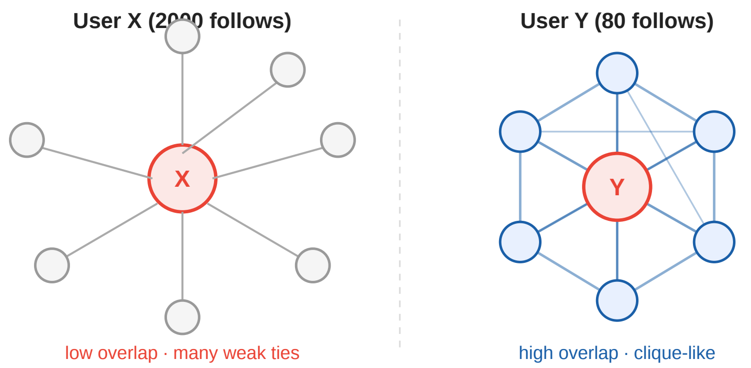
Takeaway

> The theoretical model (typed edges, STC, weak-tie theorem) cannot be inferred exactly from data, but its prediction — low overlap edges are weak — is confirmed at scale by independent empirical proxies. Model and measurement close the loop.

Quick Check

X follows 2000 accounts; most are unconnected to each other. Y follows 80 accounts; most follow each other back.

1. Using neighborhood overlap as a proxy, whose connections look more "strong-tie-like"?
2. What does the weak-tie theorem predict about the diversity of information each student receives?



Speaker notes

(1) Y has higher average overlap — dense mutual cluster, many shared neighbors. X has low average overlap — algorithmic follows create structural bridges. (2) The theorem predicts X's low-overlap edges behave like weak ties and should carry more non-redundant information. Y's high-overlap clique delivers reinforced, redundant information. The prediction matches everyday observation about echo chambers.

Quick Check — Solution

1. Y's connections look more **strong-tie-like**: dense mutual follow cluster → high average neighborhood overlap. X's large following includes many algorithm-driven follows with near-zero overlap → low average, bridge-like.

2. The weak-tie theorem predicts:

- **X** receives diverse, non-redundant information — structural signature of many weak, low-overlap edges acting as local bridges.
- **Y** receives redundant, reinforced information — structural signature of a dense clique with few outgoing shortcuts.

The prediction matches the everyday pattern of algorithm-curated feeds vs. closed-friend echo chambers.

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Wrap-Up

- **Modeling** gave us triadic closure, typed edges, STC, and the weak-tie theorem
- **Measurement** forced us to abandon the exact labeling (NP-hard) and adopt polynomial-time proxies: clustering coefficient and neighborhood overlap
- **The theorem** provided the prediction that connects the two
- **Empirical data** at scale (Onnela, Facebook, Twitter) confirm the prediction

Reading

- Chapter 3 of Easley & Kleinberg (2010).
- Kossinets & Watts, *Empirical Analysis of an Evolving Social Network*, Science **311**, 88–90 (2006) — longitudinal evidence for triadic closure on a university email network.
- Sintos & Tsaparas, *Using Strong Triadic Closure to Characterize Ties in Social Networks*, KDD 2014 — NP-hardness of MaxSTC and tractable special cases.

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Speaker notes

The wrap-up deliberately restates the modeling/measurement loop rather than listing content. Students should leave the lecture with that loop as their mental model for the rest of the course: every concept we introduce will either be a theoretical construct we cannot compute exactly, a polynomial-time proxy for such a construct, or a theoretical prediction that lets us test the proxy against data.

Image Credits & References

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